INNOVATION ECOSYSTEM REPORT

STATE OF INNOVATION HUBS IN TANZANIA (2025)

Prepared by:
Tanzania Commission for Science and Technology
(COSTECH)

1 INNOVATION ECOSYSTEM REPORT: STATE OF INNOVATION HUBS IN TANZANIA (2025)

Enhancing Sustainability, Integration, and Impact of Innovation Hubs in Tanzania's National Innovation Ecosystem

Prepared by: Tanzania Commission for Science and Technology (COSTECH)
Date: May 2025

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EXECUTIVE SUMMARY

This report presents the findings of a nationwide assessment of the state of innovation hubs in Tanzania, conducted by the Tanzania Commission for Science and Technology (COSTECH) across 25 innovation spaces in 11 regions between March and April 2025. The study aimed to evaluate the operational capacity, strategic focus, impact, and systemic challenges facing innovation hubs, with a view to strengthening their role in driving inclusive socio-economic transformation.

Innovation hubs are critical to Tanzania's broader innovation ecosystem, supporting entrepreneurship, research translation, and grassroots problem-solving. COSTECH has played a central role in shaping this ecosystem—providing technical and financial support to hubs since 2010, and fostering institutional partnerships to ensure sustainability and national alignment.

The assessment categorised findings into five key areas:

1. **Legality and Operational Structure** – 84% of hubs operate within academic institutions, benefiting from existing infrastructure but often facing bureaucratic constraints and limited autonomy. Independent hubs (16%) demonstrate flexibility but face challenges in accessing stable resources.

2. **Existing Programs and Track Record** – 90% of hubs run active incubation or acceleration programs. High-performing hubs like Twende, Kiota, Zanzibar Business Incubator, and Jamii Digital Innovation Centre have collectively supported hundreds of startups and entrepreneurs.

3. **Mentorship Capacity** – While most hubs have designated staff or focal persons, over half are overburdened with other roles. However, 76% of hubs expressed readiness to improve in mentorship, innovation management, and sustainability planning.

4. **Sustainability and Business Models** – Only 56% of hubs had documented financial sustainability plans. Most remain donor-dependent, with limited resource mobilisation capacity. Nonetheless, a few hubs have begun securing development partnerships and diversifying income.

5. **Sectoral Focus** – Hubs fall into three categories: university-linked (focused on research translation), NGO/private (grassroots innovation), and sector-specific (agritech, health, ICT, etc.).

The report also identifies **systemic challenges**, including:

- **Sustainability gaps** due to unclear budgeting and overreliance on parent institutions.
- **Underutilisation of intellectual property (IP)** within academic settings.
- **Disconnect between hubs and university research agendas**.
- **Weak resource mobilisation skills**, particularly in proposal development.
- **Low compliance with national innovation policy guidelines.**

In addition, **geographic inequity and urban bias** present significant barriers to inclusive innovation. Most hubs are concentrated in urban centres, limiting access for rural youth and women. Addressing this requires deliberate policy efforts, including transforming rural-based institutions like Folk Development Colleges into innovation nodes.

To respond to these challenges, COSTECH is rolling out a capacity-building curriculum on innovation management, IP rights, and sustainability planning—aligned with new national guidelines; The National Guideline for Innovation Spaces and the National Guideline for Spinoff companies. These interventions are designed to improve operational efficiency, foster collaboration, and equip hub managers to scale impact sustainably.

In conclusion, while Tanzania's innovation ecosystem is gaining traction, a systems-level approach is necessary for sustained transformation. With policy coherence, institutional commitment, and inclusive strategies, innovation hubs can become powerful engines for technology-driven development, job creation, and socio-economic resilience.

INTRODUCTION

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The government recognises the role that science, technology, and innovation (STI) can play in finding solutions to the development and economic challenges facing the country. The Tanzania Development Vision 2025 expresses the need to industrialise in order to create jobs for millions of young people to achieve its vision of becoming a middle-income country. Underpinning the Vision is Tanzania's 1996 National Science and Technology Policy (currently under review) which acknowledges STI tools as key for fostering industrial development and eventually developing into a modern economy.

The Tanzania Commission for Science and Technology (COSTECH) is the principal advisory organ for the Government of Tanzania on matters of Science, Technology and by extension Innovation. COSTECH is uniquely placed to support Innovation as it is a major player in both the Innovation System and Innovation Ecosystem of Tanzania.

Snapshot of the Tanzania National Innovation System

In Tanzania, the National Innovation System (NIS) is a set of distinct institutions which **jointly and individually contribute** to the development and diffusion of **new technologies** and which provides the framework within which governments form and implement policies to influence the innovation process. As such it is a system of interconnected institutions to create, store and transfer the knowledge, skills and artefacts which **define new technologies** (Metcalfe, 1995)

For Tanzania, the national innovation system is its capacity to absorb and use knowledge developed nationally and abroad.

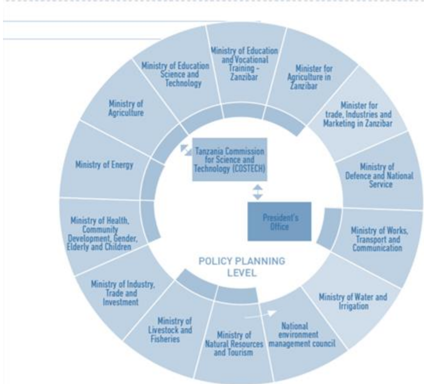
The main components of the Tanzania National Innovation System are:

i. Promotional Level
ii. Performance level
iii. Scientific and technological service level

These levels incorporate Universities, Private Sector, Industries, Government Authorities, Parastatals, Think Tanks, all the way to the President's Office. COSTECH is poised in a multi-actor, multi-sectoral apex to be able to advocate for Science, Technology and Innovation as depicted in figure 1 below. COSTECH has particularly been active in the promotion of the Triple Helix Model within the National Innovation Systems especially through the Innovative Clusters

Figure 1: 2015 NIS Report

The figure illustrates the Promotional Level of the NIS and showcases the position that COSTECH's placed in relation to the other players



Snapshot of the National Innovation Ecosystem

The term innovation ecosystem refers to the dynamic and responsive network of diverse Actors and resources that collectively drive innovation. Unlike traditional national innovation systems that rely primarily on fixed instruments and policies, an innovation ecosystem thrives on the active engagement and adaptability of its actors. These actors—such as entrepreneurs, investors, researchers, university faculty, venture capitalists, business developers, and technical service providers including accountants, designers, contract manufacturers, and providers of skills training and professional development—continuously interact, respond to emerging opportunities, and adjust to market and technological shifts to foster innovation

Figure 2: Players of the Innovation Ecosystem

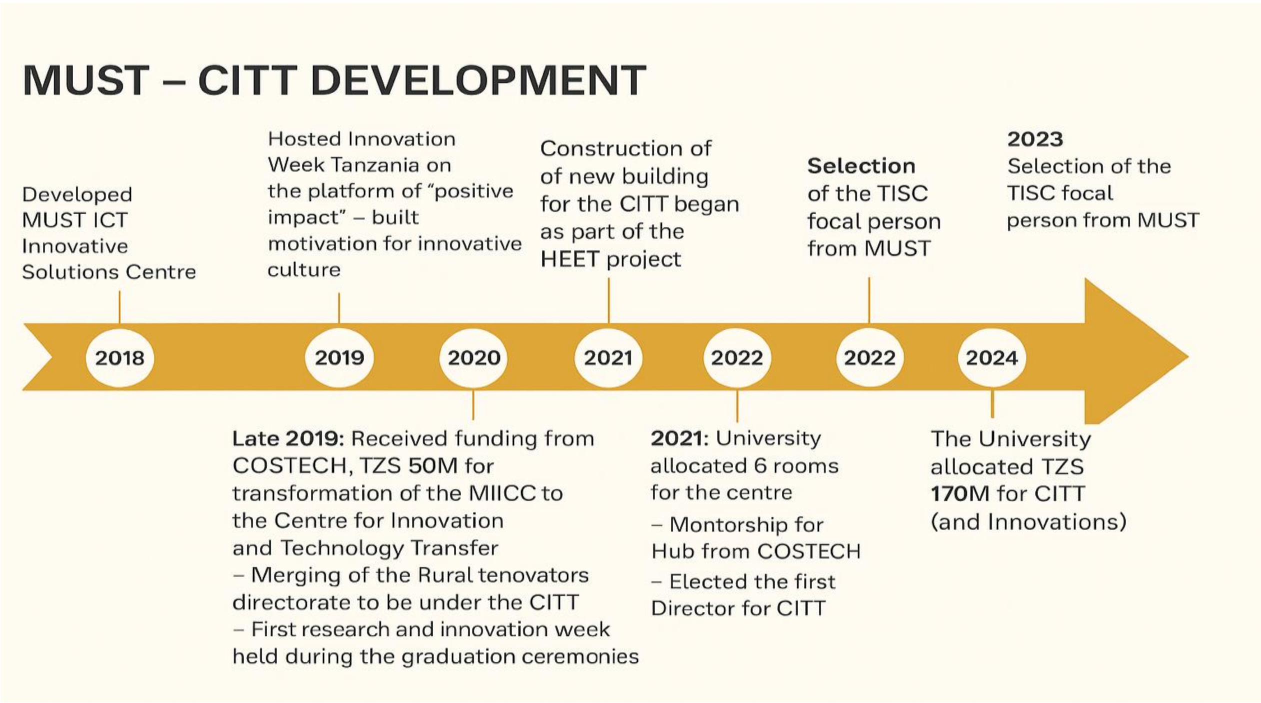
COSTECH has played an intergral role in kickstarting the Tanzania Innovation Ecosystem by setting up systems, piloting innovative programs, allowing the establishment of the first Innovation hubs (BUNI & DTBi) in the premises and setting up the first Government financing facility for Innovations. Working with partners such as the Swedish Development Agency (Sida). Over the past 14 years, COSTECH has been convening players involved in the innovation process to create the network systems needed to take innovations beyond the prototype phase and realise deep and lasting impact.

In an early-stage ecosystem such as the one in Tanzania, hubs play a crucial role in incubating and accelerating innovative ideas and building entrepreneurship skills. This is a role that COSTECH understood from early on in the efforts to build the Innovation Ecosystem. Henceforth, COSTECH has supported the establishment of innovation hubs through financial and technical support to the hubs, including linkages with Government agencies.

<u>Financial Support:</u> In 2019, COSTECH provided financial assistance of up to TZS 500 million to 15 innovation spaces to enhance their operations and sustainability. The supported 15 Innovation Hubs used the funding for infrastructure and programmes for their hubs. It led to establishment of new hubs such as Ubuntu (currently Westerwelle Haus Arusha), MIIC in MUST, SUA, UDOM, IITAF that are going strong.

<u>Technical Support:</u> In the same period, COSTECH has continued to support over 30 Innovation hubs with trainings to establish new hubs, mentorship for establishment of programmes, sustainability strategies. COSTECH has further expanded this support to 88 University hubs in East Africa Community through a collaboration with GIZ EAC.

CASE STUDY 1: How the Innovation Hub Concept can be mainstreamed in a HLI/ R&D Institution



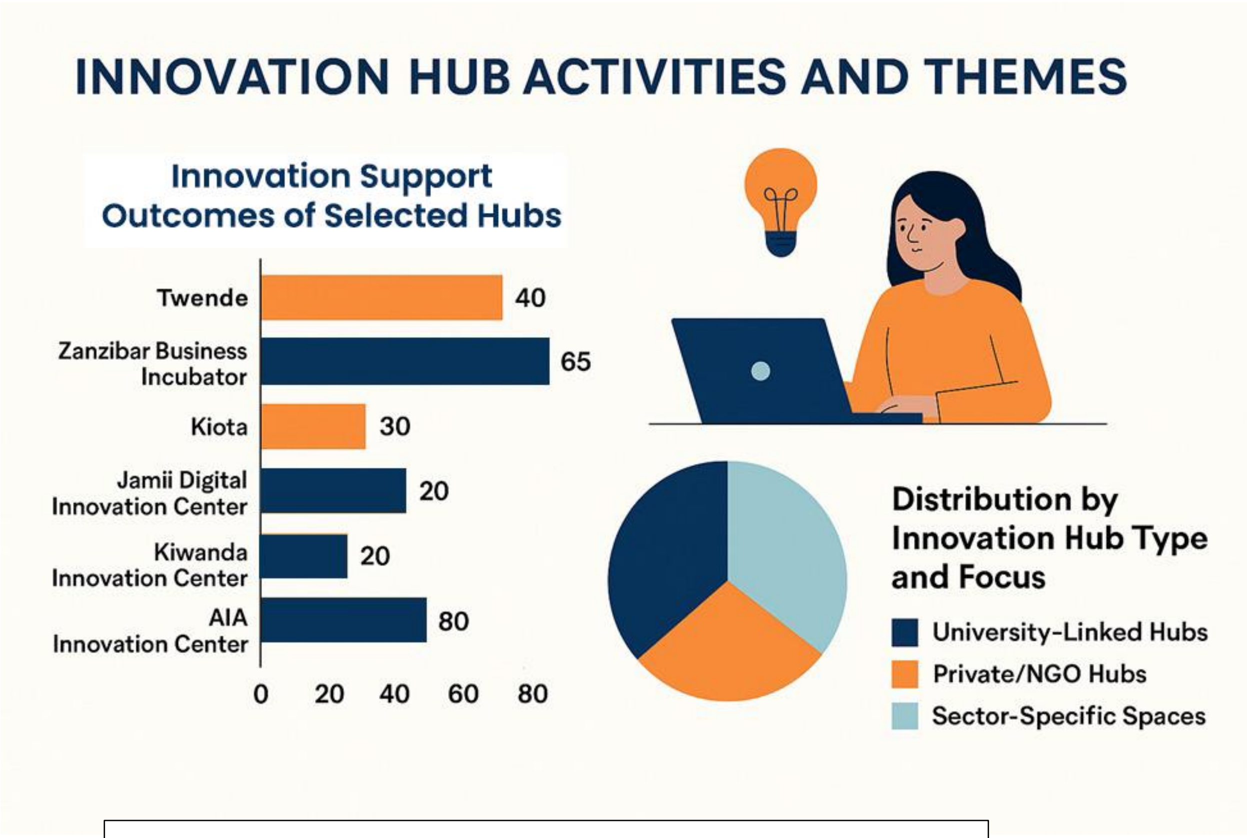
STATE OF INNOVATION HUBS IN TANZANIA IN 2025

STATE OF INNOVATION HUBS IN TANZANIA

Methodology

Over the period of 2 months; March and April 2025, COSTECH conducted a comprehensive evaluation of 25 innovation spaces across 11 regions; Iringa, Dar-es-salaam, Arusha, Tanga, Njombe, Mjini Magharibi, Mara, Morogoro, Dodoma, Mbeya, Mwanza. An online questionnaire was used, followed by key informant interviews with the co-founder(s) and in-depth interviews with key staff from selected hubs. The commission sought to establish the state of innovation hubs in Tanzania by evaluating their programmes alongside their impact in the community they serve. Further, the exercise is meant to identify key challenges faced by the spaces, and explore opportunities for further support.

This report presents the findings of the assessment, highlighting the impact of COSTECH's interventions, the associated risks, and opportunities for enhancing the sustainability, integration, and impact of innovation hubs within Tanzania's national innovation ecosystem



Numbers represent number of startups supported at the Hub

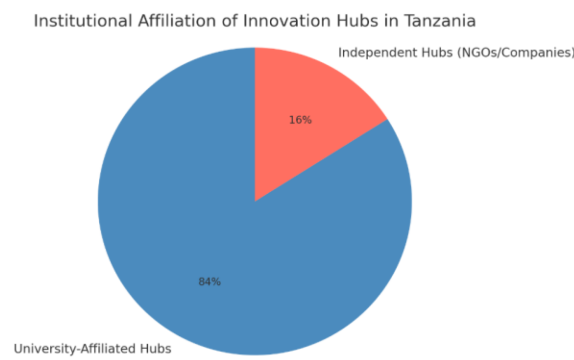
Findings

The findings of the assessment have been organized into five critical areas that collectively shape the effectiveness, sustainability, and strategic alignment of innovation hubs within Tanzania's national innovation ecosystem. Each area offers insight into both current strengths and opportunities for improvement These are the operational realities, constraints, and opportunities that define the country's emerging innovation ecosystem

A. Current Legality and Operational Structure

This dimension examines the legal status, governance frameworks, and institutional alignment of the innovation hubs. It assesses whether hubs are formally registered entities, the clarity of their mandates, and the robustness of their internal governance mechanisms. A well-defined legal and operational structure is essential for accountability, strategic partnerships, and long-term viability.

The innovation hub landscape in Tanzania is primarily anchored within academic institutions, with 84% of assessed spaces operating as affiliated entities offering legitimacy and access to existing infrastructure. However, these affiliations often come with structural limitations. Many university-based hubs lack operational autonomy, with decisions around budgeting, staffing, and programming controlled by central university authorities. This constrains their responsiveness to the fast-paced needs of innovators.



26% of the other spaces such as Twende, BongoTech, Jenga Hub, Westerwelle Haus and RLabs function independently as NGOs or companies. While nearly all hubs possess formal operational structures, those embedded within larger institutions often face staffing constraints, as core team members juggle other responsibilities. This limits their ability to fully implement innovation activities, a challenge particularly evident in cases like the UDOM Innovation Center, where team organization remains insufficiently defined. Other spaces such as Buhare CDTI, has no staff to fill the operational structure, and for that reason, one person is assigned the role of overseeing day to day activities of the space. This might have been attributed to most of the staff, who were trained in the first place, being transferred to other government institutions far from Buhare CDTI.

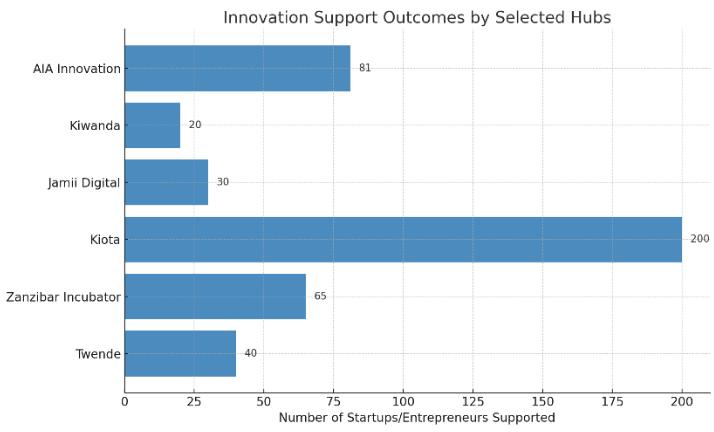
None of the identified hubs offered exclusively virtual support. All had dedicated physical spaces to host innovation activities and support startups, entrepreneurs, and innovators. However, 100% of those surveyed reported challenges related to limited workspace, which constrained their ability to effectively conduct innovation activities and scale their operations.

B. Existing Programs and Track Record

This refers to the thematic or sectoral priorities of each hub—such as ICT, agritech, health innovation, or green economy solutions. Understanding the focus area helps determine the hub's alignment with national development priorities and its potential to contribute to inclusive and sustainable growth. It also reveals how well the hub is positioned to respond to emerging trends and market demands while understanding the impact that the Hubs having been having in their communities. It further evaluates the scope, quality, and impact of programs currently being implemented by the hubs. It includes incubation and acceleration services, training initiatives, startup support, and community engagement. The track record provides evidence of past performance, credibility, and the ability to deliver measurable outcomes—key factors in attracting funding and strategic partnerships

Ninety percent (90%) of the Innovation hubs reported having active incubation or acceleration programs to support entrepreneurs and innovators in diverse sectors. Many notable startups have been nurtured by these spaces. Among the spaces with strong track records, those with a history of success in supporting innovation include Twende Social Innovation Centre, UDOM Innovation Center, Zanzibar Business Incubator, SIDO TLED, to name a few. The exception is spaces in Moduli and Buhare who currently only offer a co-working space, to name a few.

Outcome of the hubs is visible, with some notable achievements such as; 40 startups have been supported by Twende Social Innovation Centre – at least 3 of which have won MAKISATU best innovators awards, 65 startups supported by Zanzibar Business Incubator, 200 Agritech startups have been supported at Kiota, Jamii Digital Innovation Center has supported 30 entrepreneurs, Kiwanda Innovation Center supported 20 entrepreneurs and provided trainings to 100 beneficiaries, AIA Innovation Center on the other hand is currently nurturing around 81 entrepreneurs and innovators. Impact was measured for during the entire lifetime of the hub.



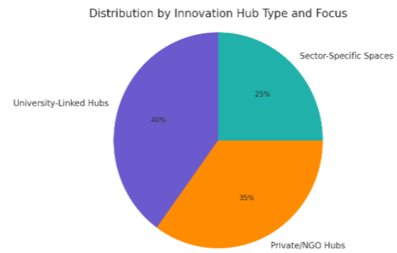
Further analysis is required to capture more nuanced outcomes of the innovation hubs, including the number of jobs created, the extent of product commercialisation, and revenue generated through community-oriented initiatives.

In terms of sector focus the hubs are grouped in four categories as indicated in the table 1 below:

Table 1: Innovation Themes and Sectoral Focus

Focus/Hub Type	Primary Innovation Focus
University-Linked Hubs	Research translation, entrepreneurship, tech transfer
Private/NGO Hubs	Grassroots innovation, social enterprise, agritech
Sector-Specific Spaces	Agriculture, digital health, manufacturing, ICT

Further, it is important to acknowledge geographic inequity, and an urban bias play a part in the impact seen in the Innovation Ecosystem. While most innovation hubs are concentrated in major cities, leaving rural and underserved regions without access to critical innovation infrastructure, mentorship, and funding opportunities. This uneven distribution reinforces socio-economic disparities, excludes grassroots innovators, and hinders the development of locally relevant solutions. For the ecosystem to drive inclusive national growth, it must intentionally expand support to peripheral areas and promote equitable access across all regions. This is discussed further in the addressing geographical inequalities sub chapter of this report.



C. Capacity for mentorship

This area focused on the human capital, institutional knowledge, and networks available within the hub to support mentorship and capacity building. It assessed the availability of experienced mentors, structured mentorship programs, and the hub's ability to facilitate peer learning and knowledge transfer. Strong mentorship capacity is a cornerstone for nurturing early-stage innovators and fostering a culture of continuous learning.

Ninety percent of the innovation hubs have either a dedicated team or a focal person responsible for managing the space and its programs. However, more than half of these reported that the team or focal person also has other primary responsibilities, which reduces their efficiency in managing the hub's programs.

Additionally, **76%** of the innovation hubs have dedicated teams that are ready and willing to receive further support in mentorship and capacity building. Priority areas identified include innovation management, project formulation, intellectual property rights (IPR), and sustainability..

D. Sustainability and Business Model

This area explored the financial and operational sustainability of the hubs. It included revenue generation strategies, funding diversification, cost structures, and long-term business planning. A resilient business model ensures that hubs can maintain operations, scale their impact, and reduce dependency on short-term donor funding. It also reflects the hub's ability to adapt to changing economic and policy environments.

This assessment also aimed to establish the state of sustainability of the innovation hubs in recognising this as an ongoing challenge. Only 56% of the assessed spaces had strategies or plans for financial sustainability, including approaches to expanding funding sources and generating revenue from the services they provide to beneficiaries or clients. This includes having documented strategic plans along with annual plans. The remaining 44% of the innovation hubs lacked clear financial sustainability strategies or plans, leaving them dependent on government funding and donor grants.

However, it was important to note that those spaces belonging to parent institutions, had strategies that were integrated into their parent institutions' plans, making independent implementation and long-term sustainability challenging. Lack of independent sustainability strategies and plans creates barriers to resource mobilization, including grants, partnerships, or corporate collaborations. To further address the issue of sustainability, some spaces for instance Zanzibar Business Incubator, AIA Innovation Centre, Monduli Digital Innovation Center and Njombe FDC Innovation Hub had taken steps to collaborate with development partners such as AfDB, JICA, KOICA, UNICEF, in the efforts to mobilize resources.

CASE STUDY 2:

The Bio-Tech Hub

Theme: Biotechnology, Health Science, Food Science

Location: Physical (Makongo, Vikindu) and eventually physical

Established: 2019 -

Sustainability Plan: build a biotechnology facility that is accessible to the general community, provide feed formulars for the community, provide extension services at a fee

Outcomes: A hub is more sustainable with an income generating facility that is dependent on the training alone.

Not all sectors require physical innovation space that is no dependent on the training alone.

Systemic Challenges

Furthermore, the assessment sort to understand the systematic challenges that faces hubs as critical engines for entrepreneurship, research translation, and grassroots problem-solving. Innovation hubs in Tanzania play a pivotal role in catalyzing development across sectors. However, the operational realities revealed through these assessments demonstrate that many of these hubs continue to struggle with entrenched systemic barriers that hinder their growth and effectiveness. These challenges are not incidental—they stem from institutional, structural, and capacity-related constraints that affect how hubs function, scale, and sustain their impact.

Therefore, we further examine five core issues confronting innovation hubs across the country: sustainability limitations, underutilization of knowledge and knowledge asset due to weak linkages with academic research, poor resource mobilization capacity, and compliance gaps with national policy frameworks. Understanding these challenges is key to unlocking the full potential of hubs as platforms for inclusive innovation and socio-economic transformation.

i. Sustainability Gaps

Challenge:
Many hubs lack independent financial planning and remain reliant on external grants and government support.

Examples:

- **University/Public-affiliated hubs**, such as UDOM Innovation Center and Buhare Digital Innovation Center, depend heavily on their parent institutions but often do not receive direct financial support. Their budgets are rarely ring-fenced, making long-term planning difficult.
- **Kiota Hub** and **Zanzibar Business Incubator**, though more proactive, have started mobilizing resources through development partners like AfDB and KOICA, highlighting potential best practices amid widespread challenges.

At IHI, we were able to transform the Hub to its own institute by clearly establishing a vision from the get-go, nurturing partnerships and tracking our impact.

Masoud Mnonji – Director, IHI

ii. Limited Intellectual Property (IP) Utilization

Challenge:
Innovative outputs from academic environments are rarely converted into protected IP assets or commercial solutions.

Examples:

- Despite being rooted in research-intensive institutions, hubs like **MUST Innovation Centre** and **SUA Innovation Hub** report low capacity in knowledge asset identification and management.
- The report emphasizes a cultural tendency for academic outputs to end with journal publications, rather than product development or commercialization efforts.

iii. Disconnect with Researchers

Challenge:
Hubs in academic settings often operate parallel to research activities, missing opportunities for integration.

Examples:

- At **UDOM Innovation Center**, team structures remain undefined, and researchers rarely utilize hub services to turn R&D results into usable interventions.
- This disconnection is common in university hubs, where innovation spaces are treated as peripheral entities, rather than strategic extensions of the research agenda.

iv. Resource Mobilization Capacity

Challenge:
Most hubs lack the internal skills and tools to design compelling, fundable project proposals.

Examples:

- The report notes widespread struggle in attracting funding, with many hubs unable to frame their programs in ways that appeal to stakeholders.
- **Jamii Digital Innovation Hub** and **Twende Social Innovation Centre** have shown initiative, but others fall behind due to low proposal development capacity and weak pitch frameworks.

v. Compliance Constraints

Challenge:

Several hubs fail to align operations with national instruments, such as the Guidelines for Innovation Spaces and the national guideline for spinoff companies.

Examples:

- University-based hubs are also affected—many struggle to integrate the guidelines into their day-to-day operations due to fragmented governance and lack of institutional ownership.

Geographic Inequity and Urban Bias in the Innovation Ecosystem

Despite the growing presence of innovation hubs across Tanzania, a significant urban bias persists in the distribution of these spaces. Most well-resourced hubs are concentrated in urban centres such as Dar es Salaam, Arusha, and Dodoma. This clustering creates geographic inequity that hinders the broader impact of the national innovation ecosystem.

► **Limited Reach to Underserved Communities:**

Rural and peri-urban regions—which make up the majority of Tanzania's population—remain largely underserved. The absence of hubs in these areas means that local innovators, particularly youth and women, lack access to critical resources such as mentorship, infrastructure, training, and markets. As a result, promising ideas in agriculture, environmental innovation, and social entrepreneurship often go unsupported or unscaled. This is why it is imperative to transform the Folk Development Colleges into centres of Innovations, making use of their unique position in the rural and peri-urban regions.

► **Reinforcement of Socio-Economic Disparities**:

When innovation support is predominantly urban, it inadvertently reinforces existing inequalities. Urban-based innovators are more likely to receive exposure, funding opportunities, and visibility through events and networks. Meanwhile, innovators in remote regions remain excluded from national and international platforms, leading to unequal development outcomes and a missed opportunity for inclusive growth.

► **Missed Context-Specific Solutions**

Local challenges require local solutions. Innovation hubs in rural areas are more likely to develop contextually appropriate technologies for agriculture, health access, clean energy, and education. Without equitable geographic distribution, the ecosystem risks becoming too detached from grassroots realities, limiting its relevance and effectiveness in addressing community-level challenges.

- **Constraints on National Development Goals**

Tanzania's development vision emphasises industrialisation, job creation, and inclusive economic participation. Geographic inequity within the innovation ecosystem undermines these goals by centralising opportunities and excluding peripheral populations. An inclusive innovation strategy must intentionally expand beyond urban zones to catalyse development at the national level.

Addressing geographic inequity and urban bias requires a deliberate policy shift. This includes incentivising the establishment of hubs in rural areas, offering subsidies or infrastructure support, and facilitating regional innovation clusters. Only by bridging this divide can the innovation ecosystem realise its full potential as a driver of inclusive and sustainable development in Tanzania.

Ongoing and Planned Interventions

COSTECH has initiated targeted capacity-building efforts for hubs. The capacity building platform will be available for hub managers to access at leisure to ensure that they can manage their duties. It will be focused on 3 areas; Innovation Management, Intellectual Property Rights, Hub Sustainability Planning.

The curriculum will be in line with the newly introduced guideline for Innovation space management by the Ministry of Education, Science and Technology. It will have the following envisioned outputs; Enhance operational efficiency, build strong institutional partnerships, Enable long-term program viability

Call to Action

Tanzania's innovation ecosystem is growing in reach, diversity, and impact. However, achieving long-term transformation demands a systems-level approach—where capacity, collaboration, commercialization, and compliance converge. With sustained investment, policy coherence, and empowered innovation leaders, these hubs can catalyze national development through technology and entrepreneurship.

Resources available

Tanzania National Innovation Framework

Map of the Innovation Ecosystem

National Guideline for Innovation Spaces

National Guideline for Spinoff Companies